

Apoorva Tummala
apoorva.tummala1234@gmail.com

11th November, 2025

Hiring Committee
College of Engineering – Robotics Department
University of Michigan
Ann Arbor, MI

Dear Hiring Committee,

I am excited to apply for the Applications Programmer Associate position at the University of Michigan's Robotics Department. With a Master's degree in Computer Science from CSU East Bay and professional experience in Python, JavaScript, SQL, and database management, I am confident in my ability to contribute to your team's development and application support initiatives.

In my role as a Systems Analyst at Michigan State University, I designed and optimized SQL tables and queries, developed over 20 Power BI dashboards, and validated API data to streamline reporting and improve operational efficiency. I have also developed web applications using Flask and Python, automated data workflows, and created documentation and training materials to support non-technical users. My experience at TCS and Aptroid Consulting strengthened my expertise in managing BI platforms, debugging code, and collaborating with cross-functional teams in higher education and enterprise environments.

I am particularly drawn to this position because of its combination of coding, application support, and collaborative problem-solving within a research-driven environment. I thrive in translating user requirements into efficient, scalable solutions, and I am eager to apply my technical skills and analytical experience to support the Robotics Department's mission while continuously improving processes.

Thank you for considering my application. I am enthusiastic about the opportunity to contribute to the Robotics Department at Michigan Engineering and would welcome the chance to discuss how my experience can benefit your team.

Warm regards,
Apoorva Tummala

APOORVA TUMMALA

apoorva.tummala1234@gmail.com | (510)714-6459 | [LinkedIn](#) | Plano, TX | Open to relocation

TECHNICAL SKILLS

- **Databases and Cloud:** MySQL, PostgreSQL, Oracle, MongoDB, AWS, Google BigQuery, Snowflake
- **Business Intelligence / Analytics Tools:** Tableau, Power BI, Qlik Sense, SAP BI4, Microsoft Excel, Google Sheets
- **Programming and ML:** Python (Pandas, NumPy, Scikit-learn, XGBoost), SQL, R, Java, C, JavaScript
- **Tools & Platforms:** SAP Ariba, Jupyter Notebook, DBeaver, Postman, Jira, ServiceNow, Git

WORK EXPERIENCE

Michigan State University (UPL) – Remote, USA

May 2024 – September 2025

Systems Analyst – Data & Procurement Systems

- Administered SAP Ariba for 40,000+ users, managing access, resolving issues, supporting system testing and enforcing governance standards to ensure data integrity and compliance across supply chain systems.
- Built 20+ Power BI and Ariba dashboards, enabling leadership and teams to track procurement KPIs and eliminating 100+ manual queries per month.
- Validated and analyzed API data with Postman, integrating external data to expand reporting coverage.
- Designed schema structures and optimized 15+ SQL tables, views, and queries for SAP Ariba–finance integration, improving reporting accuracy and reducing report preparation time from 1 hour to 10 minutes.
- Performed data profiling, cleaning and transformation, ensuring accuracy and scalable Ariba–finance integrations.
- Collaborated with cross-functional teams to configure and test system integrations, ensuring data quality assurance.
- Trained 20+ non-technical users in Ariba navigation and usage, increasing adoption and reducing support requests.
- Authored documentation and data mapping guides to standardize processes and strengthen data governance.

Technologies: SAP Ariba, Power BI, Oracle SQL, DBeaver, Postman, Jira, Kuali Financial System (KFS), UC4 (Automic)

APTIV, Tata Consultancy Services – Hyderabad, India

August 2021 – June 2022

BI Data Analyst

- Maintained Tableau & Qlik Sense BI platforms in Agile teams, handling data refreshes, backups, and system monitoring, ensuring 99% uptime for BI operations and reducing downtime by 20%.
- Developed Tableau dashboards to monitor tasks, providing real-time data visualization and improving resolution time.
- Monitored 3,000+ Tableau and Qlik Sense licenses, ensuring compliance, and reducing client licensing costs by 20%.
- Created and optimized Qlik Sense reload chains for 100+ dashboards, ensuring accurate and timely data availability.
- Processed 50+ ServiceNow tickets monthly, resolving system issues, access requests, and dashboard deployments, reducing response time by 20% and boosting user satisfaction.
- Supported BI server upgrades and documented best practices to reinforce data governance and knowledge transfer.

Technologies: Tableau, Qlik Sense, SQL, MS Excel, Qlik NPrinting, SAP BI4, ServiceNow

Aptroid Consulting Pvt Ltd – Hyderabad, India

December 2020 – July 2021

Data Support Engineer

- Extracted, transformed, and delivered data for 80+ email marketing campaigns using SQL and AWS S3, improving targeting accuracy.
- Automated repetitive data-fetching tasks with parameterized Linux scripts, reducing manual effort by 40%.
- Used advanced SQL queries (joins, unions) across multiple databases to improve campaign targeting and segmentation.
- Assisted in tasks involving Snowflake, gaining foundational understanding of data warehousing and ETL processes.

Technologies: MySQL, PostgreSQL, Linux, AWS S3, Snowflake, Jira

EDUCATION

California State University, East Bay - Hayward, CA, USA

August 2022 – December 2023

Master of Science (MS) in Computer Science

GPA: 3.86/4

Courses: Database Systems, Cloud Computing, Advanced Algorithms, System Design

Guru Nanak Institutions Technical Campus (JNTUH) - Hyderabad, India

August 2017 – July 2021

Bachelor of Technology (B. Tech) in Computer Science

GPA: 8.95/10

Courses: Artificial Intelligence, Machine Learning & Pattern Recognition, Data Warehousing, Data Mining, DB Management

CERTIFICATIONS

- Generative AI by Microsoft *July 2025*
- Google Data Analytics Specialization *December 2023*
- Basic image classification model using TensorFlow *December 2022*
- Python for Everybody Specialization *August 2020*
- Python certification from Hacker Rank *September 2020*

PROJECTS

Cloud Based Resident Management System

- Constructed a centralized system using AWS (EC2, S3, RDS), Flask, and MySQL, reducing manual data tracking by 50%.
- Provided secure resident access for updating personal information and retrieving important documents, improving user satisfaction and reducing administrative overhead.

Detecting Parkinson's Disease using Extreme Gradient Boosting

- Developed a ML model for early diagnosis of Parkinson's disease using XGBoost, achieving 94.87% accuracy.
- Preprocessed data and engineered features using Scikit-learn, NumPy, Pandas, improving model performance.

ADDITIONAL EXPERIENCE

- **Teaching Assistant for Python Lab at Cal State East Bay** *August 2023 – December 2023*
 - Taught Python to 30+ students, simplifying complex concepts for non-technical users, providing assessments, and offering detailed feedback to enhance learning and coding skills.
- **Research Assistant for Deep Learning Lab at Cal State East Bay** *February 2023 – May 2023*
 - Created and hosted a lab website with HTML, CSS, and JS (React) on GitHub and integrated RASA chatbot (local).
 - Collected and categorized 10,000+ tweets for insightful data analysis, research advancement of ongoing projects.
- **Student Internship in Cloud Computing at Red Hat Academy @ GNI** *July 2019 – August 2019*
 - Trained in scalable cloud solutions including Red Hat OpenStack and AWS, with hands-on expertise in VM setup, security, and app deployment. Proficient in deploying applications on AWS, using EC2, S3, and RDS services.
- Participated in prestigious hackathons, including Smart India (SIH), NASA Space Apps Challenge, and Hack for Humanity.
- Led a class of 70 peers by organizing study groups, coordinating events, and delegating tasks, strengthening collaboration and improving project outcomes.